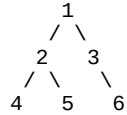


## BFS TRAVERSAL - VISUAL DIAGRAMS & REVISION SHEET

### 1. Level Order Traversal (BFS)

Tree Example:



Step-by-step BFS Queue States:

Initial:

Queue: [1]

Level 1:

Pop 1 → result: [1]

Push children: 2, 3

Queue: [2, 3]

Level 2:

Pop 2 → result: [1, 2]

Push children: 4, 5

Queue: [3, 4, 5]

Pop 3 → result: [1, 2, 3]

Push child: 6

Queue: [4, 5, 6]

Level 3:

Pop 4 → result: [1, 2, 3, 4]

Queue: [5, 6]

Pop 5 → result: [1, 2, 3, 4, 5]

Queue: [6]

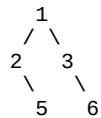
Pop 6 → result: [1, 2, 3, 4, 5, 6]

Queue: []

### 2. Right View Logic

Right view = last node of each level.

Example:



Levels:

L1: [1] → Rightmost = 1

L2: [2, 3] → Rightmost = 3

L3: [5, 6] → Rightmost = 6

Right View = [1, 3, 6]

### 3. Left View Logic

Left view = first node of each level.

L1: [1] → Leftmost = 1

L2: [2, 3] → Leftmost = 2

L3: [5,6] → Leftmost = 5

Left View = [1,2,5]

-----

#### 4. Zigzag Level Order

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Alternate between normal and reversed levels.

Levels:

L1: [1]

L2: [2,3] → reversed: [3,2]

L3: [4,5,6] → normal: [4,5,6]

Zigzag: [1,3,2,4,5,6]

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#### 5. BFS Template (must memorize)

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```
queue<TreeNode*> q;
q.push(root);

while (!q.empty()) {
    int size = q.size();
    for (int i = 0; i < size; i++) {
        TreeNode* node = q.front();
        q.pop();
        // process
        if (node->left) q.push(node->left);
        if (node->right) q.push(node->right);
    }
}
```

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#### 6. Extra BFS Style Questions

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- Bottom View
- Top View
- Vertical Order
- Reverse Level Order
- Level Averages
- Largest Value in Each Level
- Distance K
- Burning Tree

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END OF BFS VISUAL + REVISION SHEET